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Volume Comparisons in the Cerebellar Complex of Primates

I. Ventral Pons

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Abstract. Volumes of the ventral pons (VPo) were measured in 82 brains of primates, Scandentia, and Insectivora, and the relative size of the VPo was expressed by size indices. The reference line was through the prosimian average; the mean index of prosimians was set at 1. The taxonomic groups arranged in a sequence of ascending mean VPo indices were: Insectivora (0.04), prosimians (1.00), monkeys (2.55), apes (4.48) and man (9.44). Thus, the VPo of man is more than 9 times larger than that of isoponderous average prosimians, and more than 230 times larger than that of isoponderous 'basal' insectivores. The indices may reflect trends and extent of VPo evolution in primate phylogeny.

Introduction

Using a broad definition, the pons is composed of a dorsal or tegmental part (pars dorsalis pontis), containing nuclear and fiber structures associated with cranial nerves V–VIII, and a ventral part (pars basilaris pontis, basis pontis, ventral pons), containing massive longitudinal and transverse fiber systems. The ventral part constitutes the pons in a narrow sense and will be the focal point of this paper. It is mainly composed of the pontine grey matter, cortico-pontine and ponto-cerebellar fibers, and the longitudinal fibers of the pyramidal tract passing through it or closely adjacent to its dorsal boundary. The cortico-pontine fibers originate from neurons in the neocortex and synapse on the relay nuclei of the pontine grey. Postsynaptic fibers decussate predominantly within the ventral pons (VPo) and form the massive transverse fiber system that enters the contralateral cerebellar hemisphere by way of the brachium pontis or middle cerebellar peduncle.

The cerebellar hemispheres are concerned in the enormously complex organization and control of voluntary movements, whereas the cerebellar vermis is primarily related to the spinal cord functions of muscle tone and posture, and to automatic movements [Eccles, 1982]. The ventral pons can be considered a specifically mammalian configuration, and 'in the different mammalian forms the relative size of the pons seems directly proportional to

Table I. Body weights (BoW); ventral pons volumes; size indices of ventral pons (VPo), neocortex (NEO), and cerebellum (CER); ratios of VPo relative to NEO and CER; and percentage of VPo relative to brain stem

Code number	Species name	Specimens n (1)	BoW g (2)	VPo volume mm^3 (3)	VPo index (4)	NEO index (5)	CER index (6)	Ratio $\frac{VPo \cdot 100}{NEO}$ (7)	Ratio $\frac{VPo \cdot 100}{CER}$ (8)	VPo in % of brain stem (9)
Insectivora										
0577	*Tenrec ecaudatus*	1	832	2.51	0.04	0.05	0.20	0.93	0.82	0.56
0585	*Echinops telfairi*	1	87.5	0.49	0.04	0.04	0.17	0.96	0.86	0.37
Scandentia										
1263	*Tupaia glis*	2	170	13.8	0.67	0.53	0.69	1.33	3.81	2.82
1289	*Urogale everetti*	1	275	18.6	0.65	0.50	0.68	1.39	3.75	2.73
Prosimians										
3199	*Microcebus murinus*	2	54	5.9	0.61	0.82	0.94	0.80	2.58	3.09
3203	*Cheirogaleus major*	2	450	27.4	0.69	0.79	0.91	0.93	2.98	4.22
3205	*Cheirogaleus medius*	1	177	13.0	0.61	0.61	0.79	1.06	3.05	3.59
3215	*Lemur fulvus*	2	1,400	136.9	1.61	1.53	1.48	1.12	4.29	8.37
3227	*Varecia variegata*	1	3,000	168.8	1.19	1.15	1.15	1.10	4.10	6.48
3239	*Lepilemur ruficaudatus*	1	915	26.2	0.41	0.55	0.70	0.80	2.30	3.56
3243	*Avahi laniger laniger*	1	1,285	60.9	0.76	0.64	0.70	1.26	4.26	6.36
3244	*Avahi l. occidentalis*	1	860	60.3	0.99	0.77	0.85	1.36	4.56	6.60
3247	*Propithecus verreauxi*	2	3,480	169.3	1.08	0.90	0.95	1.29	4.47	7.80
3249	*Indri indri*	2	6,250	186.3	0.81	0.92	0.91	0.93	3.50	7.43
3251	*Daubentonia madagasc.*	1	2,800	304.8	2.26	1.74	1.79	1.38	4.95	11.21
3253	*Loris tardigradus*	2	322	28.9	0.91	1.18	0.87	0.82	4.13	6.00
3255	*Nycticebus coucang*	2	800	63.8	1.10	1.13	0.84	1.03	5.12	6.81
3259	*Perodicticus potto*	2	1,150	77.7	1.05	0.95	0.86	1.16	4.79	6.76
3265	*Galago senegalensis*	2	186	21.0	0.96	1.04	1.17	0.98	3.23	4.38
3267	*Otolemur crassicaudat.*	2	850	54.5	0.90	0.83	0.88	1.15	4.01	5.57
3275	*Galagoides demidoff*	2	81	13.6	1.08	1.33	1.25	0.87	3.40	4.27
3282	*Tarsius* spp.	4	125	16.5	0.98	1.12	0.96	0.93	4.01	3.93
Simians										
3287	*Callithrix jacchus*	2	280	38.8	1.34	1.61	0.98	0.89	5.40	5.95
3289	*Cebuella pygmaea*	2	120	17.5	1.07	1.52	1.00	0.75	4.21	4.59
3305	*Saguinus midas*	2	340	45.7	1.39	1.90	1.22	0.78	4.50	5.67
3311	*Saguinus oedipus*	2	380	50.9	1.44	1.77	1.04	0.86	5.45	6.38
3317	*Cebus albifrons*	1	3,100	682.2	4.72	3.41	1.96	1.47	9.49	18.74
3325	*Aotus trivirgatus*	2	830	112.1	1.88	1.77	1.16	1.13	6.36	9.28
3327	*Callicebus moloch*	2	900	105.9	1.68	1.88	0.97	0.95	6.80	7.44
3335	*Saimiri sciureus*	2	660	135.2	2.64	3.22	1.63	0.87	6.36	9.78
3341	*Pithecia monachus*	2	1,500	211.8	2.38	2.51	1.64	1.01	5.73	10.77
3363	*Alouatta seniculus*	2	6,400	394.3	1.68	1.43	0.89	1.25	7.43	13.26
3371	*Ateles geoffroyi*	1	8,000	813.3	2.98	2.76	1.68	1.15	7.00	19.70
3379	*Lagothrix lagothricha*	2	5,200	725.6	3.55	3.42	2.03	1.10	6.88	16.43
3391	*Macaca mulatta*	2	7,800	639.2	2.38	2.52	1.22	1.01	7.68	15.94
3407	*Cercocebus albigena*	2	7,900	873.3	3.23	2.70	1.43	1.27	8.86	16.32
3415	*Papio anubis*	2	25,000	2177.7	3.72	2.54	1.11	1.55	13.19	21.38

Table I (continued)

Code number	Species name	Specimens n (1)	BoW g (2)	VPo volume mm^3 (3)	VPo index (4)	NEO index (5)	CER index (6)	Ratio $\frac{VPo \cdot 100}{NEO}$ (7)	Ratio $\frac{VPo \cdot 100}{CER}$ (8)	VPo in % of brain stem (9)
3433	*Cercopithecus ascanius*	2	3,400	520.2	3.39	3.12	1.36	1.15	9.80	15.70
3453	*Cercopithecus mitis*	1	6,300	517.6	2.23	2.28	1.06	1.04	8.29	13.37
3469	*Miopithecus talapoin*	2	1,200	273.7	3.58	3.67	1.59	1.04	8.83	12.83
3473	*Erythrocebus patas*	2	7,800	814.5	3.04	3.06	1.16	1.06	10.28	16.13
3477	*Colobus badius*	2	7,000	646.4	2.59	2.17	1.26	1.27	8.08	16.22
3493	*Pygathrix nemaeus*	1	7,500	708.5	2.71	1.98	1.11	1.45	9.63	16.63
3499	*Nasalis larvatus*	1	14,000	967.4	2.44	1.68	1.11	1.54	8.68	17.69
3539	*Hylobates lar*	1	5,700	1167.9	5.38	3.22	1.98	1.77	10.70	23.94
3549	*Pan troglodytes*	1	46,000	3990.6	4.53	3.52	1.77	1.37	10.06	29.46
3551	*Gorilla gorilla*	2	105,000	5400.6	3.53	2.37	1.64	1.58	8.46	31.29
3553	*Homo sapiens*	2	65,000	10477.9	9.44	9.63	4.50	1.04	8.25	37.17

the development of neopallium and neocerebellum (i.e. roughly speaking, cerebellar hemispheres)' [Kuhlenbeck, 1975, p. 547].

From Insectivora through prosimians, monkeys and apes, and finally up to man, the neocortex is by far the most progressive structure [e.g. Stephan, 1972; Frahm et al., 1982]. Although the cerebellum is much less progressive, it, like the striatum and the diencephalon, is one of the most progressive of the non-neocortical brain components. Since both neocortex and cerebellum increase from Insectivora to man, one would expect the relay station between the two, i.e., the VPo, to increase in size as well.

Marsden and Rowland [1965] examined the size of the pons in many mammalian species, including primates. Using simple linear measurements, they found the primate pons to undergo a marked increase, reaching its maximum in man. Tilney [1927] measured the volume of the pons in various species of primates by planimetry and reported a striking evolutionary increase in size. Structural variations of the pons in several primate species were studied by Gerhard and Olszewski [1969].

In this paper allometric methods of comparison will be used, and emphasis will be placed: (1) on interspecific differences in the relative size of the VPo, its trends and intensities, and (2) on size relationships between VPo and other brain structures. Possible interrelations between relative VPo size and some components of the wide spectrum of animal behavior will be discussed in part II, which deals with the relative size of the cerebellar nuclei.

Materials and Methods

Volumes of the VPo were measured in the brains of 82 individuals of 48 species belonging to Insectivora (2 species), Scandentia (2 species), and primates (18 species of prosimians and 26 species of simians). Details are given in table I. The VPo volumes were obtained from serial sections and converted to fresh volumes. As a rule the brains were fixed in Bouin's fluid (occasionally in 10% formalin) end embedded in

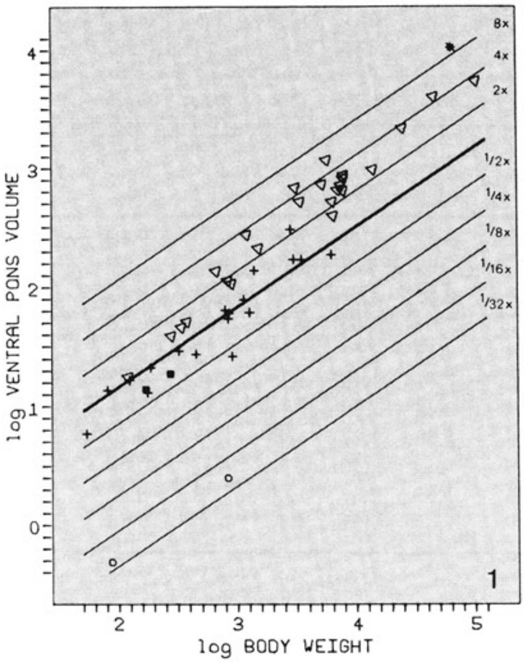


Fig. 1. Ventral pons (VPo) volume plotted against body weight (BoW) on a double logarithmic scale. The volumes are in mm^3; the weights in grams. The thick reference line is placed through the average of prosimians. Its slope is determined by least square regression analyses within primate families and subfamilies. Its equation is

$$\log VPo = -0.1795 + 0.67 \cdot \log BoW.$$

The thinner parallel lines are in distances of 0.301 (i.e. in multiples and fractions of 2) and correspond to the indices given in table I, column 4, and scaled in figures 2–4. ○ = Insectivora; ■ = Scandentia; + = prosimians; ▽ = non-human simians; ☆ = man.

paraffin. Frontal serial sections were stained with cresyl-violet for cells and by the Heidenhain-Woelcke method for fibers.

The ventral pons can easily be separated from adjacent structures. Its borders were delineated under the projection microscope from sections taken at equal intervals (5–9 from each brain). The borders were traced and the areas estimated by the Kontron-MOP-Modular-System. Volumes were calculated by applying the formula $V = A^*D/M^2$ (V: volume in mm^3; A: total area measured in mm^2; D: distance of measured sections in mm; M^2: square of linear magnification).

More details on (1) the methods of brain preparation and fixation and (2) the conversion of serial section volumes into fresh volumes are given by Stephan et al. [1981], where the fresh volumes of medulla oblongata, cerebellum (including ventral pons), mesencephalon, and neocortex are also presented. Those data were used in this study to calculate percentage relations, size indices and correlation coefficients.

For the size comparisons, three methods were used: (1) direct comparison of VPo with other structures by ratios or, when VPo was included in the reference system, by percentages; (2) interstructural regression line analyses between VPo and other structures in double logarithmic plots; and (3) size indices, which evaluate the size differences of the VPo in isoponderous species. In the third method, the VPo volumes are plotted against body weight in a double logarithmic scale, and least-square regression lines within narrowly related groups (i.e. in general, within families or subfamilies) are determined. This method is discussed in greater detail by *Stephan* et al. [1985] for the total brain. For the cerebellum, the slopes of narrowly related groups converged at 0.67. The slopes of the regression lines for the VPo (and for the cerebellar nuclei too, see part II) also converged at 0.67 and, therefore, this slope was assigned to the reference line. This line was drawn through the average of the prosimian values (n = 18). Any value on this line was considered to represent the average VPo volume of a typical prosimian. The distances from the reference line express the degree of deviation of the VPo volumes from those of a typical prosimian. The reference line itself has, of course, a distance of 0, the antilog of which is 1. Parallel lines drawn in figure 1 in multiples of 0.301 (i.e. the logarithm of 2) show when a doubling or when a half is reached.

Results

For interspecific comparisons, the size index method (regressing log VPo versus log body weight) is more appropriate; for interstructural comparisons, percentages, ratios and interstructural regressions may provide more information.

Size Indices

The ventral pons is a relay station between neocortex and cerebellum. Therefore, before dealing with the comparative size of VPo, it would be useful to have a general view of the relative size of neocortex and cerebellum in the same material as used for the VPo comparisons.

Neocortex. The volume data of the neocortex and the slope of its reference line (0.67) were taken from Frahm et al. [1982]. The reference base line of their paper was modified by shifting it from the average of 'basal' Insectivora to the average of prosimians without changing its slope in order to arrive at the same preconditions as in the VPo comparisons.

The indices of the neocortex based on the prosimian average (= 1) are given in table I, column 5, and are averaged in table II for various taxonomic groups. In general, averages of families are given, but in some cases, there were large and/or interesting differences within a given family (e.g., within Lemuridae, Lorisidae, Cercopithecidae). In these cases, averages of subfamilies are given. To avoid excessive splitting, the average indices of subfamilies within the New World monkeys (Cebidae) are not presented. There are marked differences in Cebidae too (e.g., an especially low neocorticalization in *Alouatta;* see table I). The mean values of table II are arranged in an ascending sequence, thus presenting the taxonomic units in a scale of increasing neocorticalization. This order will facilitate the recognition of similar trends in VPo indices (column 4) and cerebellum indices (column 6).

The lowest indices, indicating a neocortex of about 1/20 of that of average prosimians, were found in the 'basal' Insectivora. A low mean index (0.52) also was found in tree shrews. Tree shrews are now separated from both Insectivora and Primates and are placed in the order Scandentia [e.g. Corbet and Hill, 1980]. Within primates, the clearly lowest neocortex index was found in *Lepilemur* (0.55), and the highest index by far was found in man (9.63). Man's neocortex is about 200 times larger than that of 'basal' Insectivora and 17.5 times larger than that of *Lepilemur* of (theoretically) equal body weight. Simians (n = 26) are, on the average, 2.8 times more neocorticalized than prosimians (table II).

Cerebellum. The volume data of the cerebellum are taken from Stephan et al. [1981]. Since the data given in that paper are cerebellum + ventral pons, the VPo volumes given in table I, column 3, have been subtracted in order to arrive at the volume of the cerebellum alone.

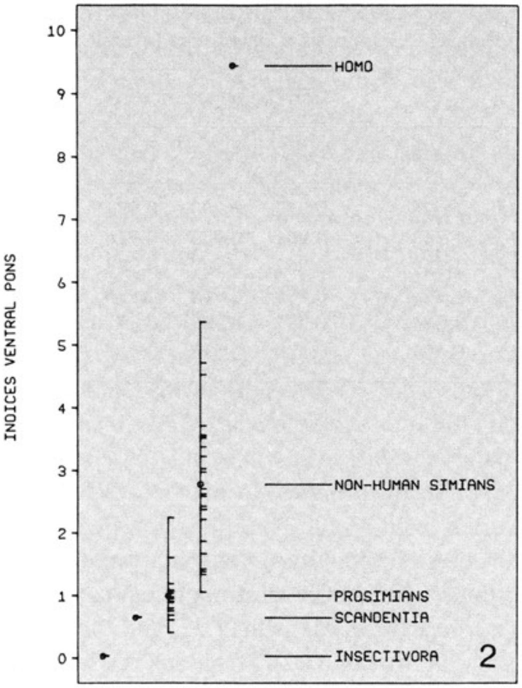
The indices of the cerebellum based on the prosimian average (= 1) are given in table I, column 6. Within primates, the range from the lowest index (*Lepilemur,* 0.70) to the highest (man, 4.50) is much smaller in the cerebellum than in the neocortex (6.4 vs. 17.5). There is the same general trend, but considerably less pronounced, from prosimians to man as found in the neocortex. The sequence of the taxonomic units for increasing cerebellum indices differs at many sites from the scale of increasing neocortex indices (table II). It differs, for example, by relatively low values for the Lorisinae, Tarsiidae and Callitrichidae and a very high value for *Daubentonia.* As in the neocortex, there is a large gap between man and other primates.

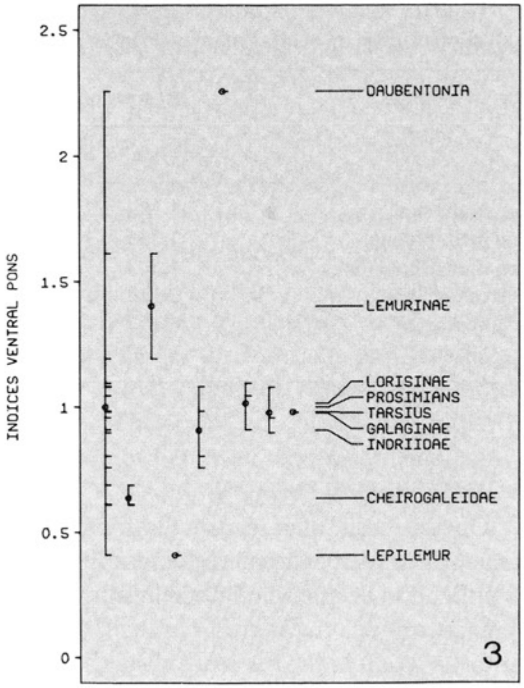
When the cerebellar indices were compared with the neocortex indices, all the simians had distinctly lower cerebellar indices, whereas in Insectivora, tree shrews and prosimians (except Lorisinae and Tarsiidae), they were similar or higher.

Table II. Means of the indices, ratios and percentages given in table I, and arranged in an ascending sequence of neocortex indices

		Number of species	VPo index	NEO index	CER index	Ratio VPo·100 / NEO	Ratio VPo·100 / CER	VPo in % of brain stem
Insectivora	0577–0585	2	0.04	0.05	0.19	0.95	0.84	0.47
Scandentia	1263–1289	2	0.66	0.52	0.69	1.36	3.78	2.78
Lepilemur	3239	1	0.41	0.55	0.70	0.80	2.30	3.56
Cheirogaleidae	3199–3205	3	0.64	0.74	0.88	0.93	2.87	3.63
Indriidae	3243–3249	4	0.91	0.81	0.85	1.21	4.20	7.05
Prosimians	3199–3282	18	1.00	1.00	1.00	1.05	3.87	5.91
Galaginae	3265–3275	3	0.98	1.07	1.10	1.00	3.55	4.74
Lorisinae	3253–3259	3	1.02	1.09	0.86	1.00	4.68	6.52
Tarsius	3282	1	0.98	1.12	0.96	0.93	4.01	3.93
Lemurinae	3215–3227	2	1.40	1.34	1.32	1.11	4.20	7.43
Callitrichidae	3287–3311	4	1.31	1.70	1.06	0.82	4.89	5.65
Daubentonia	3251	1	2.26	1.74	1.79	1.38	4.95	11.21
Colobinae	3477–3499	3	2.58	1.94	1.16	1.42	8.80	16.85
Primates	3199–3553	44	2.20	2.04	1.28	1.12	6.28	11.78
NW Simians	3287–3379	12	2.23	2.27	1.35	1.02	6.30	10.67
Monkeys	3287–3499	22	2.55	2.41	1.30	1.12	7.68	13.19
Cebidae	3317–3379	8	2.69	2.55	1.49	1.12	7.01	13.18
Simians	3287–3553	26	3.04	2.76	1.48	1.17	7.94	15.85
Cercopithecinae	3391–3473	7	3.08	2.84	1.28	1.16	9.56	15.95
Pongidae	3539–3551	3	4.48	3.04	1.80	1.57	9.74	28.23
OW Simians	3391–3553	14	3.73	3.18	1.59	1.30	9.34	20.29
Homo	3553	1	9.44	9.63	4.50	1.04	8.25	37.17

Ventral Pons. The indices of the VPo based on the prosimian average (= 1) are given in table I, column 4, and scaled in figures 2–4. The primate indices range from 0.41 in *Lepilemur* to 9.44 in *Homo.* Due to the especially low index of *Lepilemur,* the span between the lowest and highest index is larger in the VPo (1:23.0) than in the neocortex (1:17.5). As in neocortex and cerebellum, the highest index by far was found in man, but the gap between man's index and the highest indices within non-human primates was relatively small, since *Cebus* (4.72) and *Hylobates* (5.38) had very high VPo indices also. The considerable overlap between prosimian and simian indices (fig. 2) is mainly due to exceptionally high values in the prosimian aye-aye (*Daubentonia*) and in *Lemur* (fig. 3), and to very low values in Callitrichidae (fig. 4). There are strong parallels between VPo and neocortex indices. When the means of the indices (table II) are arranged in scales of increasing values, the sequence of the taxonomic units for the VPo indices is quite similar to the scale of increasing neocortex indices, but differs slightly more from a scale of increasing cerebellum indices. Similar trends and close size interrelations between VPo, neocortex and cerebellum were also reflected in the interstructural comparisons.





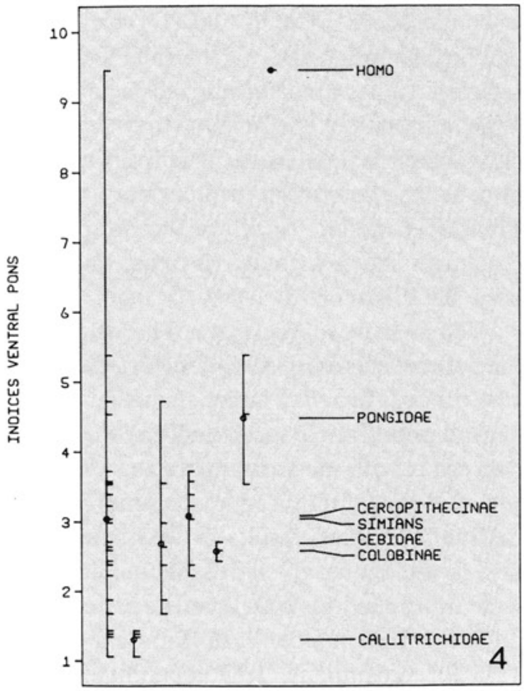


Fig. 2-4. Indices of the ventral pons. **2** Insectivora, Scandentia, prosimians, non-human simians and man. **3** Prosimians and its families and/or lower taxonomic units. **4** Non-human simians and its families and/or subfamilies. The indices are numerical expressions of the distances from the thick line given in figure 1. Each of the small horizontal bars represents the VPo index for a species. On the left, the range (vertical bars) and the mean (○) of the various taxonomic units are given, while on the right the sequences of the means are scaled.

Table III. Slopes of the interstructural regression lines and correlation coefficients in prosimians (Pros, n = 18), simians (Sim, n = 26), Primates (Prim, n = 44), and Primates + Scandentia + Insectivora (overall, n = 48)

log VPo versus	Regression line slopes				Correlation coefficients			
	Pros	Sim	Prim	overall	Pros	Sim	Prim	overall
log brain stem	1.382	1.511	1.548	1.676	0.973	0.993	0.987	0.956
log med. oblong.	1.335	1.592	1.615	1.748	0.961	0.983	0.974	0.935
log mesencephalon	1.503	1.744	1.740	1.859	0.962	0.987	0.983	0.968
log cerebellum	1.122	1.144	1.232	1.290	0.969	0.984	0.974	0.969
log neocortex	1.102	1.108	1.079	1.054	0.982	0.987	0.991	0.992
log olfactory bulb	1.040	1.261	0.419	0.199	0.580	0.577	0.057	0.009

Interstructural Comparisons

On the basis of interstructural comparisons (ratios, percentages, regression lines), it is difficult to decide whether the resulting differences are due to variation in the VPo size or to variation in the reference system, which may or may not include the VPo. The ratios and percentages are given in table I, columns 7–9; means are given in table II; correlation coefficients and slopes of the interstructural regression lines are summarized in table III. High correlation coefficients may point to close functional interrelationships, but may also be due to similar evolutionary trends. The slopes indicate which of the two plotted structures shows a greater increase and how large the difference is.

VPo Relative to Neocortex. The volume of the neocortex is from 56 times (*Hylobates*) to 134 times (*Cebuella*) larger than that of the ventral pons (ratio neocortex:VPo). Since our concern is with the variation of the VPo, the reversed ratio (VPo:neocortex) is more informative. This ratio results in very small values, so the figures given in table I, column 7, were multiplied by 100. Average ratios were 0.95 for Insectivora, 1.36 for Scandentia, 1.05 for prosimians, 1.17 for simians, and 1.12 for all primates. In man, the VPo:neocortex ratio was 1.04. The ratios were relatively uniform with no specific trends. The groups with the most strongly deviating ratios were the apes (mean 1.57) and the Callitrichidae (mean 0.82). Single species with the most strongly deviating ratios were *Hylobates,* with a maximum of 1.77, and *Cebuella,* with a minimum of 0.75. Ratios higher than 1.3 were also found in *Daubentonia,* Scandentia, and Colobinae; a ratio lower than 0.9 was found in *Lepilemur.*

In the log VPo versus log neocortex regression lines, the slopes were close to 1.1 (table III), indicating that with enlarging neocortex, the enlargement of VPo is slightly stronger. The correlation coefficients were high (r^2 = 0.98–0.99) and indicate close size relationships between VPo and neocortex.

VPo Relative to Cerebellum. The volume of the cerebellum was from 8 times (*Papio*) to 122 times (*Tenrec*) larger than that of the ventral pons (ratio cerebellum:VPo). The more informative reversed ratio (VPo:cerebellum) again resulted in very small numbers and was multiplied by 100 (table I, column 8). The ratios of Insectivora were clearly below those of primates, those of tree shrews were completely within the range of prosimians; there was little overlap between prosimians and simians. Average ratios were 0.84 for Insectivora, 3.78 for Scandentia, 3.87 for prosimians, 7.94 for simians and 6.28 for all primates. Especially high ratios were found among primates only in simians, and especially low ratios, only in prosimians. The highest ratios were found in *Papio* (13.19), followed by *Hylobates* (10.70), *Erythrocebus* (10.28), and *Pan* (10.06). The Pongidae as a group had a mean ratio of 9.74, the Cercopithecinae of 9.56, the Colobinae of 8.80, and man of 8.25. There are clear differences between Old and New World simians. For all the Old World simians, values above 8 were found (except for *Macaca,* 7.68), and for all the New World simians, values below 8 (except for *Cebus,* 9.49). The average ratio VPo:cerebellum of Old World simians was 9.34, that of New World simians 6.30. The lowest ratio was found in *Lepilemur* (2.30), and averages clearly lower than 4, in Cheirogaleidae (2.87) and in Galaginae (3.55). Within the Lemuridae, there was a clear difference between *Lepilemur* on the one hand and *Lemur* and *Varecia* on the other (2.30 vs. 4.20). From Insectivora to primates and within the primates, there is a general tendency for the VPo ratios to increase relative to the cerebellum; this trend does not, however, continue from Old World monkeys and apes to man.

The tendency of the VPo to increase more than the cerebellum results also from the log VPo versus log cerebellum regression lines, whose slopes were between 1.12 and 1.29 (table III). The correlation coefficients were high ($r^2 = 0.97$–0.98).

VPo Relative to Brain Stem. The brain stem measurements included medulla oblongata, pons and mesencephalon. Thus VPo is a constituent part of the reference system. Its percentages varied from 0.37% in *Echinops* to 37.17% in *Homo* (table I, column 9). Thus, in *Homo,* more than $1/3$ of the brain stem is VPo, whereas in *Echinops* only $1/270$ is VPo. The average percentage values, which are arranged in table II according to increasing indices of the neocortex, showed a clear trend of enlargement. The ascending sequence is broken by relatively very high average percentages for Indriidae and Colobinae and very low values for *Tarsius* and Callitrichidae. The value of the Callitrichidae is even somewhat lower than the prosimian average. With the exception of Callitrichidae, the average values for Insectivora, tree shrews, prosimian taxa, and simian taxa do not overlap (table II).

The trend of the VPo to increase much more strongly than the brain stem was also shown by the high slopes found for the interstructural log-log regression lines of these two complexes (1.38–1.68; table III). The correlation coefficients were especially high in simians ($r^2 = 0.99$), perhaps because the VPo is a constituent part of the brain stem (up to $1/3$ in man). Still higher slopes were found for the regression of the VPo with the two other components composing the brain stem. When log VPo was regressed against log medulla oblongata, slopes between 1.33 and 1.75 were found, and when regressed against log mesencephalon, even higher slopes resulted (1.50–1.86). The correlation coefficients were slightly lower than those of the brain stem as a whole but still high ($r^2 = 0.96$–0.98). The data of log VPo versus log olfactory bulb regression were added in table III in order to show that the correlation coefficients are not always high but may be distinctly lower ($r^2 = 0.01$–0.58) if there are no close functional relationships and/or similar trends.

Discussion

The VPo is a relay station between the neocortex and the cerebellum, mainly the cerebellar hemispheres, and the functional relations among these structures may be reflected in high correlations of their sizes. This is confirmed by the high correlation coefficients when log VPo is regressed versus log neocortex (table III). Similar interdependencies have been found for closely related structures of the limbic system [e.g., Stephan, 1975]. Unfortunately, no volume data for the cerebellar hemispheres are available (except for *Microcebus,* see later): the various parts of the cerebellum do not show the clear structural differences in our cell- and fiber-stained material which are needed to differentiate and delineate these complexes in histological sections. Volumes have been determined for the cerebellum as a whole [Stephan, 1972; Stephan et al., 1981], but apart from the cerebellar nuclei (part II), none of the components of the cerebellum has been measured. Therefore it would be of great benefit if the indices of the VPo and/or of the cerebellar nuclei could be used as a measure for the relative size of the cerebellar hemispheres. The validity of such an approach could be verified by macromorphological comparisons of the primate cerebellum. Such comparisons reveal fundamental trends, which have prompted Kuhlenbeck [1975] to make the general statement that VPo and cerebellar hemispheres seem to be directly proportional in relative size.

In prosimians, the two cerebellar hemispheres together appear to be similar in size to the vermis, which is very prominent. In monkeys and apes, the hemispheres clearly begin to exceed the vermis, and in man, it nearly totally disappears from the surface, since it is covered by the hemispheres. The proportion of the hemispheres now appears to be much larger than that of the vermis. According to the trends indicated by macromorphological comparisons, large cerebellar hemispheres are a progressive character, whereas a large vermis indicates a conservative cerebellum.

The results of the VPo measurements are compatible with this view. The VPo indices are low in prosimians, i.e., in the primates with the least progressive cerebellar hemispheres; they are higher in monkeys and apes and by far highest in man, which seems to have the most progressive cerebellar hemispheres of all primates. The VPo indices appear to directly reflect the size of the cerebellar hemispheres. They are very similar to the neocortex indices, and might indicate that the three structures (neocortex, VPo, and cerebellar hemispheres) are closely related in relative size [also Kuhlenbeck, 1975].

According to unpublished pilot studies, the cerebellar hemispheres of *Microcebus* are 40.7% of the total cerebellum. Assuming that (1) this value is representative of prosimians and (2) the isometric relations between VPo and hemispheres are similar to those found between VPo and neocortex (see table III), the result would be an index of about 10.1 for the cerebellar hemispheres of man and a percentage value relative to the total cerebellum of about 89%. This agrees closely with the 88% given by Eccles [1982]. The index for the rest of the cerebellum (vermis, flocculi, etc.) would be about 0.9, and such a value would indicate that, with reference to isoponderous prosimians, the relative size of the vermis in man has not changed or has decreased slightly. However, these values are deduced from unproven assumptions, and are thus highly speculative.

Additional but also indirect information on the composition of the cerebellum may be obtained from the VPo:cerebellum ratios. The average of these ratios is about 4 in prosimians and Callitrichidae, whereas in simians, it is about 8. High ratios may point to relatively large cerebellar hemispheres (lateral system, see part II), low ratios to a relatively large vermis (medial system). A tendency was found for low values in prosimians and high ones in simians, with only a slight overlap. This is another indication that the proportional size of the cerebellar hemispheres in simians is larger than in prosimians.

Besides this general trend, there are some exceptions which may be significant. Striking differences were found between New World simians (mean 6.30) and Old World simians (mean 9.34), and macromorphological inspection of the brains suggests that cerebellar hemispheres of the Old World simians are relatively larger than those of the New World simians. Especially high VPo:cerebellum ratios were found in *Papio, Hylobates, Erythrocebus,* and *Pan.* This may indicate that the lateral cerebellar system, incorporating the ventral pons, is proportionally larger in these than in other species. It is of interest that two of the forms (*Papio, Erythrocebus*), inhabit open grass lands, whereas the other two are brachiators (*Hylobates*) or semibrachiators (*Pan*). Characteristics of the motor behavior of these and other species will be discussed in part II.

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References

Corbet, G.B.; Hill, J.E.: A world list of mammalian species. British Museum (Natural History) (Comstock Publishing Associates, London 1980).

Eccles, J.C.: The future of studies on the cerebellum; in Palay, Chan-Palay, Experimental brain research, suppl. 6, The cerebellum – new vistas, pp. 607–620 (Springer, New York 1982).

Frahm, H.D.; Stephan H.; Stephan, M.: Comparison of brain structure volumes in Insectivora and Primates. I. Neocortex. J. Hirnforsch. *23:* 375–389 (1982).

Gerhard, L.; Olszewski, J.: Medulla oblongata and pons; in Hofer, Schultz, Starck, Primatologia II/2 (Karger, Basel 1969).

Kuhlenbeck, H.: The central nervous system of vertebrates. 4. Spinal cord and deuterencephalon (Karger, Basel 1975).

Marsden, C.D.; Rowland, R.: The mammalian pons, olive and pyramid. J. comp. Neurol. *124:* 175–187 (1965).

Stephan, H.: Evolution of primate brains: a comparative anatomical investigation; in Tuttle, The functional and evolutionary biology of primates, pp. 155–174 (Aldine/Atherton Inc., Chicago 1972).

Stephan, H.: Allocortex; in Bargmann, Handbuch der mikroskopischen Anatomie des Menschen, vol. 4/IX (Springer, Berlin 1975).

Stephan, H.; Baron, G.; Frahm, H.D.: Comparative size of brains and brain components; in Steklis, Comparative primate biology, vol. III, Neurosciences (Liss, New York 1985) (accepted for publication).

Stephan, H.; Frahm, H.D.; Baron, G.: New and revised data on volumes of brain structures in Insectivores and Primates. Folia primatol. *35:* 1–29 (1981).

Tilney, F.: The brain stem of Tarsius. A critical comparison with other primates. J. comp. Neurol. *43:* 371–432 (1927).

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